INTRODUCTION TO MOBILE COMPUTING

WHAT IS MOBILE COMPUTING?

Mobile Computing refers to the ability to access, process, and store information anytime and anywhere using portable computing devices such as smartphones, tablets, wearable devices, and laptops.

KEY CHARACTERISTICS OF MOBILE COMPUTING

- Portability – Devices can be carried easily.
- Connectivity – Devices connect through Wi-Fi, mobile data, Bluetooth, or NFC.
- Interactivity – Users can interact with apps in real-time.
- Context-awareness – Devices can detect location, orientation, time, and user activity.
- Mobility – Users can continue tasks even while moving.

COMPONENTS OF MOBILE COMPUTING

- Mobile Devices – smartphones, tablets, wearable tech.
- Mobile Communication – cellular networks, Wi-Fi, Bluetooth.
- Mobile Software – apps, operating systems (Android, iOS).

MOBILE APPLICATIONS

Mobile applications (mobile apps) are software programs designed to run on mobile devices.

Types of Mobile Applications

- Social Media Apps – Facebook, TikTok, Instagram
- Communication Apps – Messenger, WhatsApp
- Productivity Apps – Google Docs, Notion
- Utility Apps – Calculator, Flashlight, Weather
- E-commerce Apps – Shopee, Lazada
- Entertainment Apps – YouTube, Netflix
- Educational Apps – Google Classroom, Khan Academy
- Games – Mobile Legends, Call of Duty Mobile

CHARACTERISTICS OF MOBILE APPS

- Designed for touch input
- Optimized for small screens
- Uses device sensors (camera, GPS, accelerometer)
- Runs on mobile operating systems

NATIVE AND HYBRID APPLICATIONS

Native Applications

- These are apps developed specifically for one platform.

Examples:

- Android apps built using Kotlin or Java
- iOS apps built using Swift

NATIVE AND HYBRID APPLICATIONS

Pros:

- Fast performance
- Full access to device features
- Better user experience

Cons:

- Requires separate development for iOS and Android
- More expensive and time-consuming

NATIVE AND HYBRID APPLICATIONS

Hybrid Applications

- Hybrid apps are built using web technologies (HTML, CSS, JavaScript) and then wrapped in a native container like Cordova or Capacitor.

Examples:

- Apps built using Ionic, React Native, Flutter

MOBILE DEVELOPMENT LIFECYCLE

The mobile development lifecycle describes the step-by-step process of creating a mobile application.

PHASES OF MOBILE APP DEVELOPMENT

Planning

- Requirement gathering
- Market research
- Define features and goals

Design

- Wireframes
- UI/UX design
- User Flow Mapping

Development

- Writing code
- Implementing features
- Integrating APIs

Testing

- Functional testing
- Usability testing
- Device compatibility testing

PHASES OF MOBILE APP DEVELOPMENT

Deployment

- Publishing to Google Play or Apple App Store

Maintenance

- Updates
- Bug fixes
- Performance improvements

WHAT IS FRONT-END?

Front-end refers to the part of the application that users interact with.

- It includes everything the user sees and uses, such as:

Front-End Technologies

- HTML – structure
- CSS – design
- JavaScript – interactivity
- Frameworks – React, Vue, Angular

MOBILE APPLICATION FRONT-END WORKFLOW

This workflow describes how developers create the user interface of a mobile application.

MOBILE APPLICATION FRONT-END WORKFLOW

Requirement Analysis

- Understanding what the UI must do

Wireframing

- Low-fidelity sketches of screens

Designing (UI/UX)

- Creating polished layouts
- Using tools like Figma, Adobe XD

Coding the Interface

- Using frameworks (React Native, Flutter)
- Making screens, buttons, input forms

MOBILE APPLICATION FRONT-END WORKFLOW

Integrating APIs

- Connecting front-end to the back-end

Testing the UI

- Checking responsiveness
- Validating user interactions

Optimization

- Improving speed
- Reducing file size
- Ensuring smooth animations

Deployment

- Releasing the app to the store

THANK YOU